



Geethanjali College of Engineering and Technology (Autonomous)
Accredited by NAAC with A+ Grade; B.Tech. CSE, EEE, ECE accredited by NBA
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Question Bank: NLP (20CS32019)

PART A

UNIT I

1. Define **Morphology** in Natural Language Processing.
2. What is meant by a **morpheme**?
3. What is the difference between **free morpheme** and **bound morpheme**?
4. What is **stemming** in text processing?
5. Define **morphological analysis**.
6. What is the role of **morphological models** in NLP?
7. Explain the concept of **structure of words** in NLP.
8. Why is **document structure analysis** important in NLP?
9. What is meant by **complexity of approaches** in document structure analysis?
10. Explain the purpose of **methods used for finding the structure of documents**.

UNIT II

1. Define Syntax Analysis in Natural Language Processing.
2. What is parsing in NLP?
3. What is meant by a parse tree?
4. Define Treebank.
5. What is syntactic ambiguity?
6. What is the role of Context-Free Grammar (CFG) in parsing?
7. Explain the concept of dependency structure in syntax analysis.
8. What is meant by data-driven parsing?
9. Why are Treebanks important in NLP?
10. What is meant by multilingual NLP systems?

UNIT III

1. Define Logical Form (LF) in Natural Language Processing.

2. What is meant by semantic representation in NLP?
3. Define predicate in predicate logic.
4. What is a proposition in logical representation?
5. What is meant by quantifier in logical form?
6. What is the purpose of logical form in NLP systems?
7. Explain the concept of predicate structure in semantic analysis.
8. What is meant by encoding ambiguity in logical form?
9. Why are verbs important in logical representation of sentences?
10. What is the role of states in logical form representation?

UNIT IV

1. Name one type of cohesion that involves the use of synonyms.
2. What is **Semantic Role Labeling (SRL)**?
3. What are **core arguments**?
4. Write the purpose of **PropBank**
5. List any two **applications of semantic parsing**.
6. What are the characteristics of a good meaning representation?
7. Define **argument** with an example.
8. Discuss briefly the role of **FrameNet** in NLP
9. Define Reference Resolution in NLP.
10. What is the difference between anaphora and cataphora?

UNIT V

1. Define a **Language Model**.
2. What is meant by an **n-gram model**?
3. State the **Markov assumption** used in n-gram models.
4. What is **perplexity** in language model evaluation?
5. What is the purpose of **smoothing** in language models?
6. Write the difference between **unigram and bigram models**.
7. What is **Maximum Likelihood Estimation (MLE)** in language modeling?
8. What is meant by **Out-of-Vocabulary (OOV) words**?
9. What is a **Neural Network Language Model (NNLM)**?
10. What is **language model adaptation**?

PART B

UNIT I

1. Explain the concept of **morphology in Natural Language Processing** and describe the structure of words with suitable examples.

2. Discuss **words and their components** such as roots, prefixes, and suffixes with examples.
3. Explain the **issues and challenges in morphological analysis** in Natural Language Processing.
4. Describe different **morphological models used in NLP** and explain their significance.
5. Explain different **methods used for finding the structure of documents** in NLP.
6. Demonstrate how **morphological analysis can be performed for a given set of words** using suitable examples.
7. Apply a **morphological model to analyze the structure of words** in a given sentence.
8. Demonstrate the process of **document structure analysis** using appropriate methods and examples.
9. Apply different **approaches for identifying document structure** and explain their working with examples.
10. **Analyze the complexity and performance of different approaches used for document structure analysis** and compare their advantages and limitations.

UNIT II

1. Explain the process of parsing natural language and discuss its importance in Natural Language Processing.
2. Describe Treebanks as a data-driven approach to syntax and explain how they are used in NLP applications.
3. Explain different representations of syntactic structure used in NLP with suitable examples.
4. Explain the concept of phrase structure grammar and illustrate the construction of a parse tree for a sample sentence.
5. Construct a dependency structure for a given sentence and explain the dependency relations between words.
6. Demonstrate the Top-Down parsing algorithm using an example sentence and grammar rules.
7. Demonstrate the Bottom-Up parsing algorithm with a suitable example and explain each step in the parsing process.
8. Apply the CKY (Cocke–Kasami–Younger) parsing algorithm to parse a given sentence using a CNF grammar.
9. Explain the multilingual issues in NLP and discuss challenges in developing multilingual parsing systems.
10. Analyze different models used for ambiguity resolution in parsing and explain how they help in resolving syntactic ambiguity.

UNIT III

1. Explain the concept of Logical Form (LF) and discuss its role in semantic representation in NLP.
2. Describe the process of encoding ambiguity in logical form with suitable examples.

3. Explain the representation of verbs in logical form with examples.
4. Explain the concept of states in logical form representation with suitable illustrations.
5. Construct the logical form representation for a given sentence and explain the predicate structure involved.
6. Demonstrate how predicate logic is used to represent the meaning of sentences in NLP.
7. Construct predicate structures for different types of sentences and explain their components.
8. Represent the logical form of a complex sentence containing verbs and states using predicate logic.
9. Explain how logical form helps in semantic interpretation of natural language sentences with examples.
10. Analyze different sources of ambiguity in logical form representation and discuss techniques used to resolve them.

UNIT IV

1. Compare **FrameNet** and **PropBank** in terms of their **semantic representation, annotation approach, and applications in NLP systems**.
2. Analyze how **Semantic Role Labeling (SRL)** improves **meaning extraction and understanding** in Natural Language Processing applications.
3. Explain the concept of **Predicate–Argument Structure in NLP** and illustrate how **predicates and their arguments are identified in a sentence** with suitable examples.
4. Consider the sentence: “*The professor explained semantic parsing to the students.*” Identify the **predicate and arguments**, represent the sentence using **Predicate–Argument Structure notation**, and explain each **argument role**.
5. Analyze how **frame-based semantic representation** captures sentence meaning and explain the **components of a semantic frame** used in NLP systems.
6. Explain the concept of **meaning representation in NLP systems** and discuss the **cohesion structure** with suitable examples.
7. In the sentence “*The cat chased the mouse through the garden.*”, identify the **predicate–argument structure** and assign appropriate **semantic roles** (Agent, Theme, Location, etc.) with justification.
8. Explain the concept of **discourse in NLP**, its importance in language understanding, and identify the **head and morphological type** (NP, VP, AP, AdvP) of the following segments:
 - The president of the company
 - Looked up the chimney
 - Angry as a hippo
 - Rapidly like a bat
9. Describe the **semantic resources used in shallow semantic parsing** and explain the working principles of **FrameNet and PropBank** with suitable examples.
10. Explain **Reference Resolution in NLP** and discuss different **reference resolution algorithms** with suitable examples.

UNIT V

1. Explain the **n-gram language model** with a suitable example.
2. Illustrate how **unigram, bigram, and trigram models** compute sentence probability.
3. Explain the **evaluation of language models** using coverage rate and perplexity.
4. Discuss **Maximum Likelihood Estimation (MLE)** and its limitations in language modeling.
5. Explain different **smoothing techniques** used in language models.
6. Describe the **Kneser–Ney smoothing method** and its advantages.
7. Explain **Neural Network Language Models (NNLM)** with architecture.
8. Analyze the challenges involved in **large-scale language modeling**.
9. Compare **class-based language models and variable-length language models**.
10. Explain **multilingual and cross-lingual language modeling approaches**.